



Mohamed Saber Abbas Elsayed

Computer & Systems Engineering Student · Frontend Developer

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github.com/mhmdsabeer2029 | LeetCode: [MohamedSaber123](#) | HackerRank: [mhmdsabeer2029](#)

PROFILE

Computer and Systems Engineering student at Alexandria University with a cumulative GPA of 3.97/4.0 and a portfolio spanning low-level C systems, modern web applications, and AI-integrated products. Currently training under a Senior Frontend Engineer at Capgemini to refine production-grade frontend engineering practices. Strong fundamentals in algorithms and data structures, with proven Hard-tier problem-solving on LeetCode. Builder mindset across the stack, with shipped academic, client, and personal projects.

EDUCATION

B.Sc. in Computer and Systems Engineering

Sep 2023 – Jun 2028 (Expected)

Faculty of Engineering, Alexandria University — Alexandria, Egypt

- Cumulative GPA: **3.97 / 4.0**.
- Core coursework: Programming I & II, Data Structures, Algorithms, Computer Architecture, Digital Logic Design, Information Theory, Signals and Systems.

EXPERIENCE

Frontend Engineering Trainee

Apr 2026 – Present

Mentorship under **Eng. Amr Matar**, Senior Frontend Engineer, **Capgemini** — Remote · Paid

- Following a structured, mentor-led curriculum to adopt a senior-level approach to modern frontend engineering.
- Practising production-grade patterns in JavaScript, component architecture, performance tuning, and UI engineering.
- Delivering project iterations per training cycle and receiving direct technical review on code quality, structure, and design decisions.

TECHNICAL SKILLS

- **Languages:** C, Python, JavaScript (ES6+), HTML5, CSS3, SQL.
- **Frontend & UI:** Vanilla JS engineering, React, Tailwind CSS, Responsive Design, Canvas API, Accessibility.
- **Backend & Tooling:** FastAPI, Node.js, REST APIs, Docker, n8n, Git, GitHub, GNU Make.
- **Systems & Algorithms:** Data Structures, Algorithms, SDL2, Pygame, Information Theory, Digital Logic.
- **AI & Computer Vision:** Groq LLM API, MediaPipe, KaTeX, LLM application design, Prompt engineering.
- **Practices:** Clean Architecture, Code Review, Performance Tuning, Cross-platform Build, Technical Writing.

SELECTED PROJECTS

ScriptFlow — Premium SaaS Landing Platform

HTML5, CSS3, Vanilla JS, Canvas API

Live: mhmdsabeer2029.github.io/ScriptFlow · Source: github.com/mhmdsabeer2029/ScriptFlow

- Engineered a four-page conversion-optimized SaaS marketing site with zero external dependencies.
- Built an interactive HTML5 particle hero, a custom smart cursor, scroll-driven reveals via IntersectionObserver, and an animated billing toggle.
- Designed a luxurious dark-navy and gold visual system using CSS custom properties, fluid typography, and inline SVGs.

JARVIS — Conversational AI Assistant

Python, FastAPI, Groq API, JavaScript, KaTeX

Live: dist-lykkvdkb.devinapps.com

- Built a full-stack AI assistant integrating the Groq LLM API with a FastAPI backend and a streaming frontend.
- Implemented native LaTeX rendering with KaTeX and Computer Modern typography for mathematics-quality output.
- Engineered a ChatGPT-style multi-conversation workspace with sidebar search, pinning, and persistent state.
- Tuned the streaming pipeline to first-token latency of approximately 500 milliseconds.

Professional Chess Engine

C, SDL2, SDL_mixer, GNU Make

Source: github.com/mhmdsabeer2029/Chess_by_C

- Engineered a 100% FIDE-compliant chess engine and graphical user interface from the ground up in C.
- Implemented castling, en passant, pawn promotion, threefold repetition, insufficient material, checkmate, and stalemate.
- Built a three-layer modular architecture providing O(1) check detection and clean separation between logic and rendering.
- Integrated Forsyth-Edwards Notation (FEN) serialization for game persistence and validation.

Agent Istanbul — Cross-Border Business Platform

HTML5, CSS3, JavaScript, Responsive Design

Live: agent-istanbul.com

- Designed and engineered a live multi-service marketing platform for a Turkish trade and consulting agency.
- Delivered hero, about, features, process, statistics, testimonials, and blog modules with a tightly tuned responsive layout.

Nokhbat — UpSkill Academy

Web, PHP, JavaScript, Responsive UI

Live: nokhbat.ayaxd.com

- Built a complete e-learning storefront covering eight knowledge tracks with course listings, ratings, lifetime-access bundles, instructor highlights, and a polished conversion flow.

Neural Hand Tracking System

MediaPipe, JavaScript, Canvas, WebRTC

- Implemented real-time hand-landmark detection using Google MediaPipe Hands.
- Rendered a 21-point hand skeleton over live video, classified finger states and gestures, exposed confidence telemetry, maintained gesture history, and integrated an AI analysis pane.

Wordle Game and AI Solver

Python, Pygame, Algorithms

Source: github.com/mhmdsabeer2029/World-Project

- Implemented a complete Wordle clone in Pygame with color-coded feedback and correct duplicate-letter handling.
- Engineered an entropy-aware solver combining letter-frequency analysis with positional scoring, averaging four guesses (worst case six).

Quran Web Application

HTML5, CSS3, Vanilla JS, SPA

- Built a feature-rich single-page Quran reader with typography-first design, search, navigation, and a companion engineering-law revision module.

Gmail Deadline Tracker (n8n Automation)

n8n, Docker, Google Sheets API, Gmail API

- Self-hosted workflow that reads a live Google Sheet of academic assignments, computes time-to-deadline, and dispatches Gmail reminders on a precise schedule.

DEFLATE Compressor — LZ77 Stage

Python, Information Theory

Source: github.com/mhmdsabeer2029/Signals_project

- Implemented Stage 1 of a four-stage simplified DEFLATE compressor with a 32 KB sliding-window encoder, hash-accelerated match lookup, and a lossless byte-level decoder.

8-bit Arithmetic Logic Unit

Logisim, Digital Logic

- Designed and verified a gate-level ALU in Logisim with arithmetic and bitwise operations, status flags, and control-driven operation select.

ACHIEVEMENTS

- Solved Hard-tier problems on LeetCode (handle: [MohamedSaber123](#)).
- Active competitive programmer on HackerRank (handle: [mhmdsabeer2029](#)).
- Cumulative GPA 3.97 / 4.0 at Alexandria University, Faculty of Engineering.

LANGUAGES

- **Arabic** — Native.
- **English** — C1+.
- **French** — A1.